

METHOD STATEMENT SUBMISSION FORM (MSSF)

To	The <i>Project Manager's</i> Delegate	Attn.	Mr. Peter Poon
Ref. No.	CFB/MSSF/PM/2026/000198	MSSF No.	MSSF/000010/A
Subject	Method Statement for Concrete Remedial Works Inside Cavern		
Item	Description		
1.	We hereby submit the captioned method statement for your acceptance.		
Remarks :			
Expected Reply Date : 15 May 2026			
From : The <i>Contractor's</i> Representative Name : Hung Ling Fung, Samuel Signature :  Date : 25 Apr 2026			

Method Statement for Concrete Remedial Works Inside Cavern

Revision No.: A

Position	Name	Signature	Date
Drafted by: Assistant Engineer	Yu Cho Yiu, Curtis		25 Apr 26
Reviewed by: Safety Manager	Tsang Chuen Ming, Michael		25 Apr 26
Checked by: Site Agent	Hung Ling Fung, Samuel		25 Apr 26
Endorsed by: Construction Manager	Yeung Lung Fung, Kenneth		25 Apr 26

Content

1. INTRODUCTION	2
2. METHOD OF CONSTRUCTION.....	3
3. REPAIR CLASSIFICATION OF CONCRETE WORK.....	4
4. PLANT AND EQUIPMENT.....	7
5. SAFETY.....	7
6. ENVIRONMENTAL.....	8

APPENDIX A –INSPECTION AND TEST PLAN

APPENDIX B – RISK ASSESSMENT

REVISION

ISSUE	REVISION NO.	DATE OF ISSUED	DESCRIPTION
1 st submission	A	25/4/2026	1 st submission to the PM

1. Introduction

This method statement outlines the detailed procedures for structural works, including rebar fixing, formwork erection, concrete pouring in sequences, inspection and testing plans (ITP), and safety risk assessments (RA) related to the concrete remedial works as part of the C4 contract.

Following the completion of concreting, this document will provide an in-depth description of the concrete remedial methodology, emphasizing safety and compliance with the stringent quality control standards mandated by Contract No. DC/2025/07 – Relocation of Sha Tin Sewage Treatment Works to Caverns – Civil Works for Sewage Treatment Facilities and Associated Works.

The principle methods outlined in the subsequent sections will be evaluated based on the actual site conditions and may be modified as necessary. Any amendments to this method statement will be submitted to the *Project Manager* for review and acceptance.

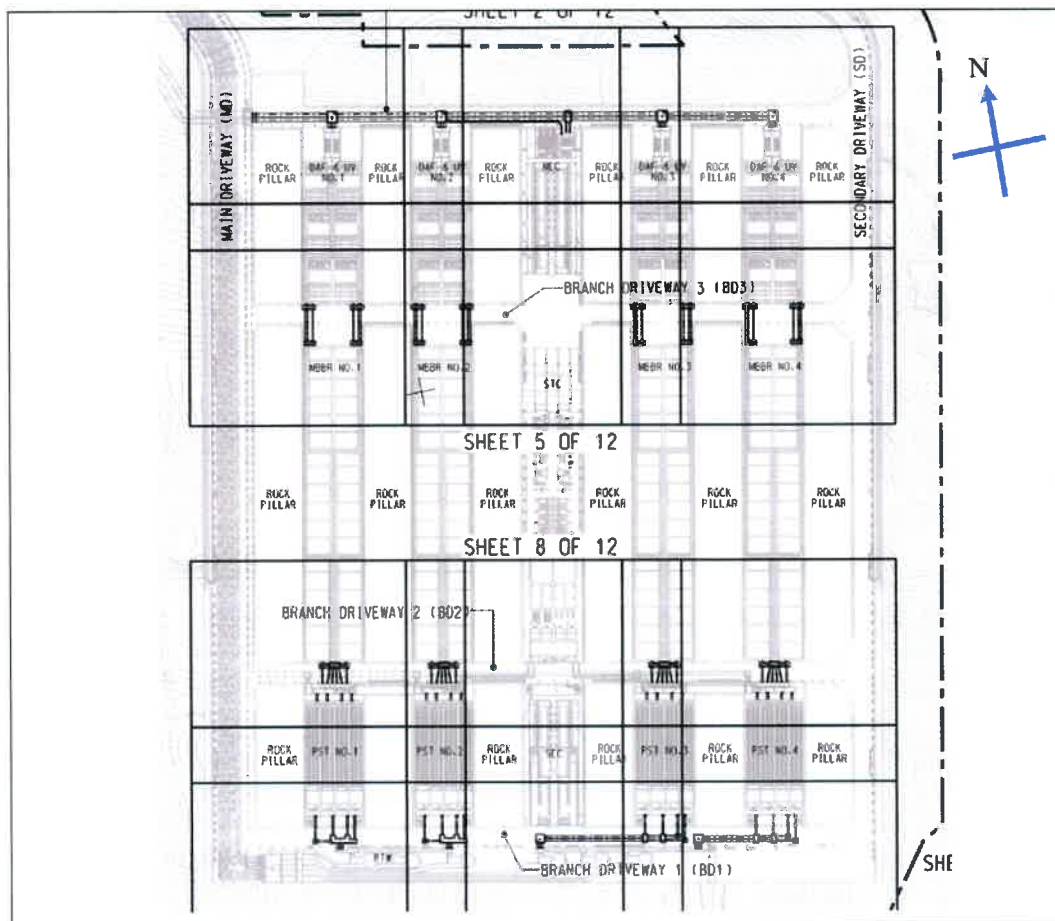
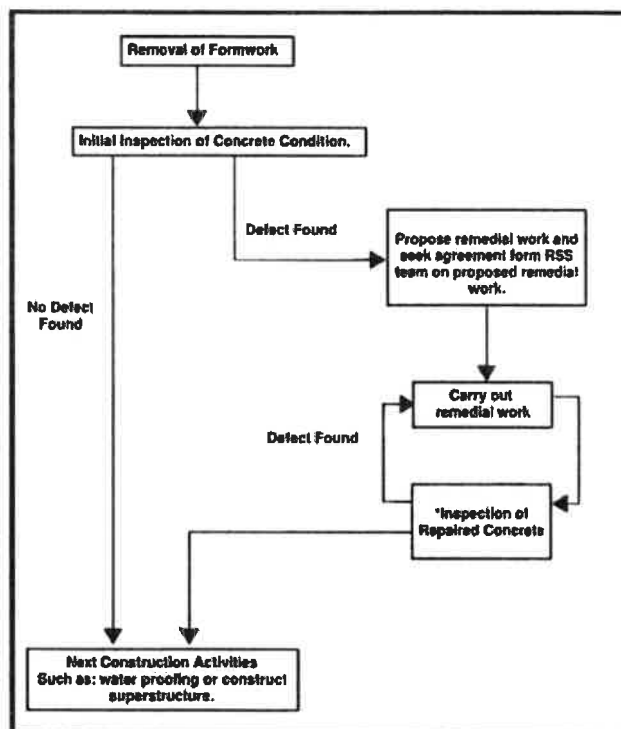


Figure 1. Plan of Cavern Complex

2. METHOD OF CONSTRUCTION

2.1 Flow Chart of Concrete Remedial Inspection stated as below:



2.2 Conduct a thorough post-pour inspection in collaboration with the RSS team to identify any defects in the reinforced concrete. Document the findings with photographs and detailed notes, specifying the location and nature of each defect.

2.3 Clearly mark the identified defective areas by spraying paint or markers on site for visibility. While ensure that markings are easily recognizable to all team members and indicate the extent of the area requiring repair.

2.4 Develop a comprehensive plan for the remedial work, including materials, methods, and timeline and seek formal agreement from the RSS team on the proposed remedial work to ensure alignment and approval before proceeding.

There are normal concrete issues with remedial proposal are mentioned in **Section 3**.

2.5 Implement the agreed-upon remedial actions, ensuring adherence to safety guidelines and construction standards. And utilize appropriate materials and techniques to ensure the integrity and durability of the repairs.

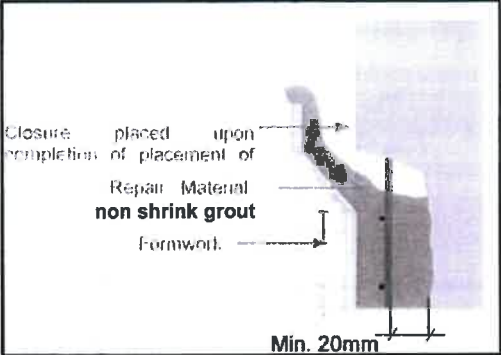
2.6 Conduct a detailed inspection of the repaired areas to ensure that all defects have been adequately addressed and document the inspection results before next activities work proceeding.

3. REPAIR CLASSIFICATION of CONCRETE WORK

In this section, we will outline several common concrete issues along with their respective solutions.

No	Defect	Remedial work	Classification
1	Non-uniform /non-smooth surface /step joint / Gap	1.Hand chipping loose material / foreign 2.Mechanical grinding the surface. 3.Rubbing concrete surface with sand paper. 4.Apply repair mortar. 5.Smooth the repaired area with trowel	Minor
2	Grout run	1.Saw cut the concrete around the defective area10mm depth. 2.Roughen the surface by light scrubbling 3.Break out the concrete behind the rebar 20 mm depth when loose material behind exposed rebar. 4.Clean the surface to remove dust loose and unsound material by blow machine. 5.Soaked repair zone with clean water. 6.Apply non-shrink grout according to manufacturer's recommendation.	Minor
3	Blow hole	1. Roughen the surface by light scrubbling. 2.Clean the surface to remove dust loose and unsound material. 3.Soaked repair zone with clean water. 4. Fill up the blow hole with Trowel able mix repair mortar.	Minor
4	Insertion of foreign Material into concrete	1. Saw cut the concrete around the defective area10mm depth.	Minor

		2. Break out the concrete around the foreign material. 3. Remove the foreign material. 4. Break out the concrete behind the rebar 20 mm depth when loose material behind exposed rebar. 5. Clean the concrete surface thoroughly, removes any dirt, loosen concrete or debris. 6. Apply non-shrink grout, according to manufacturer's recommendation. 7. Smooth the repaired area with trowel.	
5	Missing Chamfer	1. Set out the chamfer line on the concrete wall or slab. 2. Saw cut the corner of the concrete wall or slab along the setting out line to form the chamfer. 3. Smooth the surface of the chamfer with electrical grinding machine.	Minor
6	Honeycomb concrete extends to a depth 5-75mm when rebar is not encountered.	1. Saw cut the perimeter of the defective area to a depth of 5mm below concrete surface. 2. Breaking out the defective concrete with portable electric chisel /breaker. 3. Removal all loose material and carry out surface preparation. 4. Mixing of Non shrink grout in proportion as recommended by manufacturer, then using trowel or other suitable placing tools to transport the repair material to the prepared substrate. The repair material pressed into the substrate to develop intimate contact without void. 5. Polyester mortar be protected from damage during hardening, and be covered with polythene sheeting to ensure that the hardening process is not unduly prolonged.	Minor
7	Honeycomb concrete extend to a depth >75mm , area <300mm x 300mm (rebar exposed - Dry Packing Repair)	1. Saw cut the perimeter of the defective area to a depth of 5 mm below concrete surface. 2. Breaking out the defective concrete with portable electric chisel /breaker to expose 10mm underneath reinforcement. 3. Apply zinc rich primer to reinforcement. 4. Removal all loose material and carry out surface preparation as per manufacturer's recommendation of Non shrink grout 5. Mixing of non shrink grout in proportion of Trowelable as per manufacturer's recommendation, then transport to a confined space and compacted with rodding tools (hardwood) to produce a dense repair material by trowel. 6. Polyester mortar be protected from damage during hardening, and be covered with polythene sheeting to ensure that the hardening process is not unduly prolonged. Repair Material Non shrink grout Rodding Tools (hardwood) Min. 20mm Dry Packing Repair	Minor

8	Honeycomb concrete extend to a depth >75mm , area>300mmx300mm(rebar exposed-Form and Grout)	1.Saw cut the perimeter of the defective area to a depth of 5mm below concrete surface. 2. Breaking out the defective concrete with portable electric chisel /breaker to expose 10mm underneath reinforcement. 3. Apply zinc rich primer to reinforcement. 4. Removal all loose material and carry out surface preparation as per manufacturer's recommendation of non shrink grout . 5. Pre-prepare letter box shutters (formwork) and then mixing of Non-Shrink Grout in proportion of pourable/flowable as per manufacturer's recommendation. 6. Polyester mortar be protected from damage during hardening, and be covered with polythene sheeting to ensure that the hardening process is not unduly prolonged.  <i>Form and Grout</i>	Major
9	Crack ($\geq 0.1\text{mm}$ <0.25mm)	1. Identify the location and extent of crack. 2. V-cut approx. 5x10-15mm deep along the cracks. 3. Remove all loose particles and clean the inside surfaces of the cracks. 4. Infill the V-cut areas with Non-Shrink Grout up to the desire profile.	Minor
10	Crack(>0.25mm)	1. identify and agree the crack (>0.25mm wide) to be injected. 2. Clean and remove all loose materials along the crack with appropriate tools, the crack shall be cleaned by blowing out with clean compressed air. 3. Install injection nipple along the cracks about specialist recommendation 200-300mm interval and seal up the crack surface with epoxy paste. 4. Inject the grouting material of epoxy resin starting from the lowest end of the crack. 5. Stop inject while grouting material come out from the next nipple. 6. Inject the grouting materials to next nipple same as above until the injection work complete 7. Remove the nipples next day after work completion.	Major
11	Tie bolt hole	1 Remove debris, loose material, or any contaminants from the tie bolt hole. 2. Using the hand tool to mark the hole to ensure a good bond with the repair materials. 3. Apply zinc rich primer to tie bolt nut (if necessary) 4. Apply bonding agent to the concrete surface before apply remedial grout. 5. Apply non-shrink grout to fill the tie bolt hole.	

4. PLANT AND EQUIPMENT

All plants and equipment will be inspected prior to the mobilization on site to ensure that they are in good working condition and comply with the current regulations.

The trade of workers, major plants and equipment will be deployed to carry out the works are as follows:

Trade of Workers	Quantity
Concreter Breaker	1 (As required)
Plasterer / Skilled Labour	1 – 2 (As required)

Plant / Equipment	Quantity
Electrical Breaker	2
Electrical Drill	2
Steel Trowel	4
Electrical Circular Saw	1

Materials	Quantity
Non shrink Grout	As required
Zinc rich prime	As required
Timber Shutter	As required
Bonding agent	As required

5. SAFETY

- 5.1. The latest approved project safety plan should be referred for all general safety consideration and the risk for each particular activity to be carried out should be thoroughly considered.
- 5.2. Site safety training and pre-work workshop would be conducted prior the operation. The risk assessment is attached in **Appendix E**.
- 5.3. Working area should be fenced off to prevent unauthorized entry.

- 5.4. Regular checks on all plant and machinery will be performed and recorded on Form 1 and valid copies of Form 3, 4 and 5 will be maintained.
- 5.5. Relevant Permit – to - work (i.e. permit – to - dig, permit – to – lift, etc.) will be implemented.

6. ENVIRONMENTAL

The latest updated environmental management plan should be referred.

6.1. Working Hours

The working hours for the proposed works will be from 07:00 to 19:00, except Sundays, Public holidays and those non-working days instructed by the Engineer, unless otherwise allowed by the Construction Noise Permit granted in the future.

6.2. Dust Control

It is anticipated that the proposed works will involve small quantities of earthwork. The temporary stockpile will be covered with tarpaulin sheeting. Sprayed water will be used to suppress dust if necessary.

6.3. Waste / Construction Refuse

Construction refuse will be collected on a regular basis. Construction debris will be taken away by grab lorries to a designated dumping site stated in the Waste Management section of the Environmental Management Plan (EMP). The construction and demolition waste will be disposed offsite under the trip tickets system and prescribed site procedures according to EMP will be strictly followed. Construction waste will require proper sorting before leaving the construction site.

6.4. Water Pollution

The potential impacts on water quality may come from the surface run-off from the works as well as the plant cleaning water to the surface drainages. The concerned surface run-off will be pump out to the sedimentation tank. Ditches will be dug around the works area to collect spillage of water and discharge to a sedimentation tank. Desilting tank will be provided to remove suspended solids before discharging to public drain upon obtaining the discharge license.

6.5. Noise mitigation measures

Noise mitigation measures will be implemented following the latest updated environmental management plan. Mitigation measures will be provided for the noisy activities, such as breaking of hardtop and rock masses. Conduct noise monitoring to ensure noise level is controlled with 75dB.

Contract No. DC/2025/07

Relocation of Sha Tin Sewage Treatment Works to Caverns – Civil Works for Sewage Treatment Facilities and Associated Works
Method Statement for Concrete Remedial Works Inside Cavern

Appendix A –INSPECTION AND TEST PLAN

Ref.no. CFB/ITP/0010

Method Statement for Concrete Remedial Works Inside Cavern

Date: 25 Apr 2026

								Inspection Representative		
Item	Activity	Inspection / test method	Sepecification(if applicable)	Acceptance Criteria	Inspection Frequency	Verifying Record / Document	Inspected by / Tested by (Position)	S/C	ASTECJV	PM
A	Preparation Works									
A.1	Material Submission	Document Review		As stated in Particular Specification/ General Specification	Once	Contractor's submission form (CSF) Method Statement and Notice of No Objection (NNO)	RSS	H	H	✓
B	Post Concreting									
B.1	Post Pour Surface Checking	Visual Inspection	N/A	As defined in specification/ drawing	After concreting/ prior to handover to client	RISC form	RSS	H	H	✓
B.2	Repaired defect inspection	Visual Inspection	N/A	As defined in specification/ drawing	After repair	RISC form	RSS	H	H	✓

Remarks: H - Hold Point C - Control Point W - Witness

Contract No. DC/2025/07

Relocation of Sha Tin Sewage Treatment Works to Caverns – Civil Works for Sewage Treatment Facilities and Associated Works

Method Statement for Concrete Remedial Works Inside Cavern

Appendix B – RISK ASSESSMENT

Risk Assessment - Method Statement for Concrete Remedial Works Inside Cavern

No	Task	Hazards	Related Parties	Likelihood	Severity	Risk Rating	Control Measures	Likelihood Residue	Severity Residue	Risk Residue	Action By	Monitor & Review By	PPE
1.	Site Preparation	Injury by vehicle or machinery	Workers	2	3	VH	- Signaller should be provided for the vehicle or machinery working area. - Fenced off the machinery or vehicle moving area if possible. - All workers should wear reflect clothing. - All vehicle should be installed reversing alarm. - Do not work close to the vehicle or machinery moving area.	1	2	VL	E / SE / F / SS	CM / SA / GF / SO	1 / 5 / 8
2.	Formwork Erection	Back Injury	Workers	3	1	L	- Training for proper manual handling posture should be provided to all relevant workers. - Adjustable working platform should be used to prevent excessive body stretching. - Manual handling assessment should be conducted to assess the hazard. - Re-arrange the work to prevent a single worker to perform the task. - Equipment such as chain block should be considered.	2	1	VL	E / SE / F / SS	CM / SA / GF / SO	1 / 5 / 8
		Person Falling from Height	Worker	2	3	VH	- Ensure that working platforms with guardrails, toe-boards complying with the statutory requirements are used by workers. - Visual check the working platform by an appointed competent person at least every 7 days - Make sure that valid statutory certificates Form 5 is available. - For those locations that working platforms are not available, Independent lifeline, fall arrestor and safety harness must be provided. Strictly enforce to use the PPE. - Provide safety training to workers.	1	2	VL	E / SE / F / SS	CM / SA / GF / SO	1 / 4/5/ 7/8/ 14/15
		Hands were Being Trapped by the Formwork.	Workers	2	1	VL	- Communicate with relevant parties about the operation procedures. - Adequate supervision should be arranged to supervise the work. - Only authorized workers were allowed to get close to the work area. Effective communication should be maintained amongst the working group. - Safety training should be provided to relevant workers	1	2	VL	E / SE / F / SS	CM / SA / GF / SO	1 / 5 / 8/ 14

Risk Assessment - Method Statement for Concrete Remedial Works Inside Cavern

No	Task	Hazards	Related Parties	Likelihood	Severity	Risk Rating	Control Measures	Likelihood Residue	Severity Residue	Risk Residue	Action By	Monitor & Review By	PPE
		Failure of LG	Plant Operator/ Workers	2	3	VH	➤ Ensure examined by RPE and possess valid statutory certificates for the lifting gears. ➤ Check the lifting gears before use and ensure that they are good condition & safe for use. ➤ Proper colour coding should be provided. ➤ Make sure that does not exceed the SWL during operation. ➤ Provide suitable training for safe use of LG to users	1	2	VL	E / SE / F / SS	CM / SA / GF / SO	1 / 5 / 8 /
		Collapse of Crane Lorry/Hydraulic Crane	Plant Operator/ Workers	2	3	VH	➤ Ensure examined by RPE and possess valid statutory certificate ➤ Ensure the crane lorry/ hydraulic crane is placed in an even position with all outriggers fully extended. ➤ Visual check by an appointed competent person at least every 7 days. The crane lorry / hydraulic crane must submit the valid certificate to the Safety Department to obtain the Plant Permit issued by site. ➤ Provide adequate training of safe use of machine to operators / workers. ➤ Communicate with relevant parties about the operation procedures. Appoint a competent operator with appropriate training to operate the machine	1	1	VL	E / SE / F / SS	CM / SA / GF / SO	1 / 5 / 8
		Tripping	All person on site	2	1	VL	➤ Proper materials stacking should be enforced. ➤ Demarcated materials/ tools storage area with warning sign displayed. Communicate with relevant parties about the material stacking position	1	1	VL	E / SE / F / SS	CM / SA / GF / SO	1 / 5 / 8
		Hit by Crane Lorry	Workers	1	3	H	➤ Only authorized workers should be allowed to stay near the assembly area. ➤ Adequate lighting should be maintained at the works area. ➤ The working area should be properly fenced off with warning sign prominently displayed. ➤ Traffic controller/ banksman should be assigned to control the movement of the crane lorry ➤ Provide safety training to all workers. ➤ All workers must wear reflective vest in order to alert drivers.	1	2	VL	E / SE / F / SS	CM / SA / GF / SO	1 / 5 / 8

Risk Assessment - Method Statement for Concrete Remedial Works Inside Cavern

No	Task	Hazards	Related Parties	Likelihood	Severity	Risk Rating	Control Measures	Likelihood Residue	Severity Residue	Risk Residue	Action By	Monitor & Review By	PPE
		Falling Objects	All person	1	3	H	➤ No materials should be on the formwork whilst lifting. ➤ Secure/ remove loose parts. ➤ Make sure that all area near the working position should be fenced off with suitable warning notice displayed. ➤ Keep the workplaces clean & tidy ➤ Safe working procedure should be implemented. ➤ Safety training should be provided to relevant workers. ➤ Communicate with relevant parties about the operation procedures.	1	2	VL	E / SE / F / SS	CM / SA / GF / SO	1 / 5 / 8
3.	Repairing the defective concrete	Person fall from a height	All person	3	2	H	➤ Ensure that working platforms with guardrails, toe-boards complying with the statutory requirements are used by workers. ➤ Visual check the working platform by an appointed competent person at least every 7 days ➤ Make sure that valid statutory certificates Form 5 is available. ➤ For those locations that working platforms are not available, Independent lifeline, fall arrestor and safety harness must be provided. Strictly enforce to use the PPE. ➤ Provide safety training to workers.	1	2	VL	E / SE / F / SS	CM / SA / GF / SO	1 / 5 / 8
		Collapse of platform	Workers	1	3	H	➤ Appoint a competent worker to erect the platform. ➤ Appoint a competent person to inspect the scaffold and issue the statutory Form 5 stating the inspection result. ➤ Make sure that the scaffold was erected in compliance with the COP. ➤ For non-typical shape of scaffold, appoint competent engineer to design the scaffold & check the compliance during the erection process & shall be counter-checked by the ICE after completion ➤ Daily check the scaffolding & ensure that they are safe & in good condition. ➤ Suitable training should be provided to workers.	1	1	VL	SE / E / GF / F / SO / SS	CM / GF / SM / SO	1 / 5 / 8

Risk Assessment - Method Statement for Concrete Remedial Works Inside Cavern

No	Task	Hazards	Related Parties	Likelihood	Severity	Risk Rating	Control Measures	Likelihood Residue	Severity Residue	Risk Residue	Action By	Monitor & Review By	PPE
		Eye injury with broken concrete dust flying into eye	Workers	2	2	M	- Suitable eye protector should be provided - Keep the workplaces clean and tidy - Water suppression should be adopted - Safe working procedure should be implemented - No person, except the involved workers should be allowed to get close to the working position - Safety training should be provided to relevant workers	1	2	VL	SE / E / GF / F / SO / SS	CM / GF / SM / SO	1 / 5 / 8 / 13
		Back injury caused by awkward position during works	Workers	3	2	M	- Training for proper manual handling posture should be provided to all relevant workers. - Manual handling assessment should be conducted to assess the hazard - Rearrange the work to ensure sufficient resting time being allocated to relevant workers - The workplace should be kept clean and tidy all time - A clear and unobstructed access should be maintained	1	2	VL	SE / E / GF / F / SO / SS	CM / GF / SM / SO	1 / 5 / 8
		Hand injury by hand tools	Workers	2	2	M	- Provide gloves to worker - Sufficient lighting should be provided at the position of work - Safety training should be provided to relevant workers	1	2	VL	SE / E / GF / F / SO / SS	CM / GF / SM / SO	1 / 5 / 8
		Irritation by grout	Workers	2	2	M	- Provide protective rubber gloves to involved worker when using chemicals - Conduct health risk assessment base on the MSDS - Keep the workplaces clean and tidy - Safe working procedure should be implemented - No person except the involved workers should be allowed to get close to the working position - Safety training should be provided to relevant workers	1	2	VL	SE / E / GF / F / SO / SS	CM / GF / SM / SO	1 / 5 / 8 / 10
		Eye injury during grouting	Workers	2	2	M	- Suitable eye protector should be provided - Conduct health risk assessment base on the MSDS - Keep the workplaces clean and tidy - Safe working procedure should be implemented - No person except the involved workers should be allowed to get close to the working position	1	2	VL	SE / E / GF / F / SO / SS	CM / GF / SM / SO	1 / 5 / 8 / 13

Risk Assessment - Method Statement for Concrete Remedial Works Inside Cavern

No	Task	Hazards	Related Parties	Likelihood	Severity	Risk Rating	Control Measures	Likelihood Residue	Severity Residue	Risk Residue	Action By	Monitor & Review By	PPE
4.	Material lifting to Installation (Plywood, timber and etc.)	Hit by Crane / Materials	Workers	3	2	H	➤ A well-trained banksman shall be assigned to control the movement of the crane lorry / plants. ➤ The lifting area should be properly fenced off with warning sign prominently displayed. ➤ All workers stay close to the crane lorry must wear reflective vest in order to alert plant operators or drivers. ➤ Only authorized workers should be allowed to stay near the path of the crane lorry. ➤ Guide rope should be used to control the movement of the materials whilst lifting. ➤ Provide safety training to all workers.	1	2	VL	E / SE / F / SS	CM / GF / SM / SO	1 / 5 / 8
		Collapse of Crane	Plant Operator / Rigger	3	2	H	➤ All crane lorry must submit the valid certificate to the Safety Department to obtain the Plant Permit issued by site. ➤ Ensure examined by RPE and possess valid statutory certificate. ➤ Appoint a competent operator with appropriate training to operate the machine. ➤ Visual check with an appointed competent person at least every 7 days. ➤ Provide adequate training of safe use of machine to operators / workers. ➤ Communicate with relevant parties about the operation procedures. ➤ Ensure the crane is placed on an even ground with fully extended outriggers.	1	2	VL	CM / E / GF / F / SO / SS	SA / GF / SM / SO	1 / 5 / 8
		Failure of LG	Plant Operator / Rigger	3	2	H	➤ Check the lifting gears before use and ensure that they are good condition & safe for use. ➤ Ensure examined by RPE and possess valid statutory certificate. ➤ Proper colour coding should be provided. ➤ Make sure that does not exceed the SWL during operation. ➤ Provide suitable training for safe use of LG to users.	1	2	VL	CM / E / GF / F / SO / SS	SA / GF / SM / SO	1 / 5 / 8

Risk Assessment - Method Statement for Concrete Remedial Works Inside Cavern

No	Task	Hazards	Related Parties	Likelihood	Severity	Risk Rating	Control Measures	Likelihood Residue	Severity Residue	Risk Residue	Action By	Monitor & Review By	PPE
		Falling Object during lifting	All personnel on site	2	2	M	➤ All loosen materials on the lifting object should be removed before lifting ➤ The rigger method should be safe and completed by competent rigger ➤ Make sure that all area near the working position should be fenced off with suitable warning notice displayed. ➤ Nobody is allowed to stand underneath a lifting object ➤ Keep the workplaces clean & tidy ➤ Safe working procedure should be implemented. ➤ Safety training should be provided to relevant workers.	1	2	VL	SA / CM / GF / F / SS	SA / GF / SO	1 / 5 / 8 / 15
5.	Erection/ Dismantling of scaffold	Falling Objects	All person on site	2	3	VH	➤ Wearing of safety helmet on site should be enforced ➤ Secure/ remove loose parts before transportation ➤ Keep the workplaces clean & tidy ➤ Materials should not be stacked to close to the edge of metal scaffolding or working platform. ➤ Loosen materials should be stored in materials cage or bucket. ➤ Not used materials should be secured. ➤ Keep and maintain the toe-boards in good condition. ➤ Provide suitable training for safe stacking & rigging of materials to workers ➤ Hand held tools should be held by tag rope where possible release from worker hand.	1	2	VL	SE / E / GF / F / SO / SS	GF / SM / SO	1 / 5 / 8
		Fall of Person	Workers	2	3	VH	➤ Working platform with rigid fencings and toe-boards are erected for workers to use. ➤ For those locations where rigid guardrails are not available, independent lifeline, fall arrestor and safety harness must be provided. ➤ Strictly enforce to use the PPE. ➤ Code of practice for erecting metal scaffolding should be fully complied. ➤ Provide safety training to workers.	1	2	VL	SE / E / GF / F / SO / SS	CM / GF / SM / SO	1 / 4 / 5 / 7 / 8 / 14 / 15

Risk Assessment - Method Statement for Concrete Remedial Works Inside Cavern

No	Task	Hazards	Related Parties	Likelihood	Severity	Risk Rating	Control Measures	Likelihood Residue	Severity Residue	Risk Residue	Action By	Monitor & Review By	PPE
		Collapse of platform	Workers	1	3	H	➤ Appoint a competent worker to erect the platform. ➤ Appoint a competent person to inspect the scaffold and issue the statutory Form 5 stating the inspection result. ➤ Make sure that the scaffold was erected in compliance with the COP. ➤ For non-typical shape of scaffold, appoint competent engineer to design the scaffold & check the compliance during the erection process & shall be counter-checked by the ICE after completion ➤ Daily check the scaffolding & ensure that they are safe & in good condition. ➤ Suitable training should be provided to workers.	1	1	VL	SE / E / GF / F / SO / SS	CM / GF / SM / SO	1 / 5 / 8
		Back Injury	Workers	2	1	VL	➤ Training for proper manual handling posture should be provided to all relevant workers. ➤ Manual handling assessment should be conducted to assess the hazard. ➤ Provide suitable & adequate machines to assist handling if possible.	1	1	VL	SE / E / GF / F / SO / SS	CM / GF / SM / SO	1 / 5 / 8
6..	Concreting	Dust	All person on site	3	1	L	➤ The working position should be enclosed. ➤ Safety training should be provided to relevant workers. ➤ Water suppression should be considered.	2	1	VL	E / SE / F / SS	CM / SA / GF / SO	1 / 2 / 5 / 8 / 13
		Hit by concreting pipes	Workers	11	2	VL	➤ Provide safety training to workers.	1	1	VL	E / SE / F / SS	CM / SA / GF / SO	1 / 5 / 8
		Eye injury with concrete materials blowing into eye	Workers	3	1	L	➤ Keep the workplaces clean & tidy ➤ Safe working procedure should be implemented. ➤ No person, except the involved workers should be allowed to get close to the working position. ➤ Safety training should be provided to relevant workers	2	1	VL	E / SE / F / SS	CM / SA / GF / SO	1 / 5 / 8 / 13

Risk Assessment - Method Statement for Concrete Remedial Works Inside Cavern

No	Task	Hazards	Related Parties	Likelihood	Severity	Risk Rating	Control Measures	Likelihood Residue	Severity Residue	Risk Residue	Action By	Monitor & Review By	PPE
		Fall from Height	Workers	2	2	M	➤ Ensure the working platforms used by workers are fitted with guardrails and toe-boards. ➤ Visual check the mobile working platform by an appointed competent person at least every 7 days. ➤ For those locations that working platforms are not available, independent lifeline, fall arrestor and safety harness must be provided. ➤ Regular check the condition of the mobile working platform and make sure that valid Form 1 had been issued. ➤ Provide & maintain safe access to the workplaces and enforce to use Strictly enforce to use the PPE. ➤ Weekly visual checking of the anchorage of the independent lifeline, fall arrestor and full- body harness should be conducted by appointed site engineers/ supervisors	1	2	VL	SE/E/ GF/F/ SO/ SS	CM/GF / SM/SO	1/5 /8
		Skin burn due to contact of concrete	Workers	2	2	M	➤ Wear alkali-resistant gloves (e.g., nitrile or rubber gloves) to protect hands from direct contact with wet concrete. ➤ Wear knee-high waterproof boots when working in or around wet concrete to protect feet and lower legs. ➤ Wash off any concrete splashes immediately with clean water to prevent prolonged exposure. ➤ Use tools (e.g., trowels, screeds) to handle wet concrete instead of hands whenever possible.	1	2	VL	E / F SS	CM / SA / GF / SO	1/5/10/16
7.	Loading and unloading the materials (formwork) by crane lorry	Hit by Crane / Materials	Workers	3	2	H	➤ A well-trained banksman shall be assigned to control the movement of the crane lorry / plants. ➤ The lifting area should be properly fenced off with warning sign prominently displayed. ➤ All workers stay close to the crane lorry must wear reflective vest in order to alert plant operators or drivers. ➤ Only authorized workers should be allowed to stay near the path of the crane lorry. ➤ Guide rope should be used to control the movement of the materials whilst lifting. ➤ Provide safety training to all workers.	1	2	VL	E / SE/ F /SS	CM / GF / SM / SO	1 / 5/ 8

Risk Assessment - Method Statement for Concrete Remedial Works Inside Cavern

No	Task	Hazards	Related Parties	Likelihood	Severity	Risk Rating	Control Measures	Likelihood Residue	Severity Residue	Risk Residue	Action By	Monitor & Review By	PPE
		Collapse of Crane	Plant Operator / Rigger	3	2	H	➤ All crane lorry must submit the valid certificate to the Safety Department to obtain the Plant Permit issued by site. ➤ Ensure examined by RPE and possess valid statutory certificate. ➤ Appoint a competent operator with appropriate training to operate the machine. ➤ Visual check with an appointed competent person at least every 7 days. ➤ Provide adequate training of safe use of machine to operators / workers. ➤ Communicate with relevant parties about the operation procedures. ➤ Ensure the crane is placed on an even ground with fully extended outriggers.	1	2	VL	CM / E / GF / F / SO / SS	SA / GF / SM / SO	1 / 5 / 8
		Failure of LG	Plant Operator / Rigger	3	2	H	➤ Check the lifting gears before use and ensure that they are good condition & safe for use. ➤ Ensure examined by RPE and possess valid statutory certificate. ➤ Proper colour coding should be provided. ➤ Make sure that does not exceed the SWL during operation. ➤ Provide suitable training for safe use of LG to users.	1	2	VL	CM / E / GF / F / SO / SS	SA / GF / SM / SO	1 / 5 / 8
		Falling Object during lifting	All personnel on site	2	2	M	➤ All loosen materials on the lifting object should be removed before lifting ➤ The rigger method should be safe and completed by competent rigger ➤ Make sure that all area near the working position should be fenced off with suitable warning notice displayed. ➤ Nobody is allowed to stand underneath a lifting object ➤ Keep the workplaces clean & tidy ➤ Safe working procedure should be implemented. ➤ Safety training should be provided to relevant workers.	1	2	VL	SA / CM / GF / F / SS	SA / GF / SO	1 / 5 / 8 / 15

Risk Assessment - Method Statement for Concrete Remedial Works Inside Cavern

No	Task	Hazards	Related Parties	Likelihood	Severity	Risk Rating	Control Measures	Likelihood Residue	Severity Residue	Risk Residue	Action By	Monitor & Review By	PPE
8.	Dismantling of formwork	Back Injury Caused by Material Handling	Workers	2	2	M	- Manual handling assessment should be conducted to access the hazard. - Re-arrange the work to prevent a single worker to perform the task. - Equipment such as pulley should be provided in order to reduce the transport distance. - A clear & unobstructed access should be maintained.	1	1	VL	E / F / SS	CM / GF / SA / SO	1 / 5 / 8
		Falling Object	All person on site	3	2	H	- Make sure that all area near the working position should be fenced off with suitable warning notice displayed. - Secure / remove loose parts. - Keep the workplaces clean & tidy. - Safe working procedure should be implemented. - Safety training should be provided to relevant workers. - Communicate with relevant parties about the operation procedures.	2	1	VL	SE / E / GF / F / SO / SS	CM / GF / SM / SO	1 / 5 / 8
9.	Working within Confined Space	Hazardous gas, vapor, dust or fume, or deficiency of oxygen present	Workers	2	3	VH	- Use water jet to clean the surface of the pipeline to remove sludge or other deposits. - Check the air by competent person - Monitor the air in the confined space continuously until everyone has left the confined space - Strictly comply with and implement the Confined Space Work-Permit system. - Worker should hold a valid certificate of "Certified Workers". - Adopt safe working procedures in accordance with the COP. - Provide ventilation system to supply fresh air - Fence off the working area and display warning notice. - Provision of non-sparking tools and specially protected lighting. - Safety training and training about the emergency procedures for fire should be provided to relevant workers. - Full body protective equipment (eg. face mask with cartridge filter, eye protector,) should be worn to all relevant workers to avoid inhale or contact with hazardous gas.	1	2	VL	E / F / SS	CM / SA / GF / SO	1 / 5 / 8 / 11 / 12

Risk Assessment - Method Statement for Concrete Remedial Works Inside Cavern

No	Task	Hazards	Related Parties	Likelihood	Severity	Risk Rating	Control Measures	Likelihood Residue	Severity Residue	Risk Residue	Action By	Monitor & Review By	PPE
		Ingress of hazardous gas vapour, dust or fume to the confined space	Workers	2	3	VH	➤ Use blowers to direct fresh air through the air duct and remove air impurities. ➤ Use appropriate gas detectors (i.e. explosion-proof design) to monitor the air in the confined space continuously until everyone has left the confined space	1	2	VL	E / F SS	CM / SA / GF / SO	1 / 5 / 8 / 11 / 12
		sludge or other deposits being present that are liable to give off hazardous gas, vapour, dust or fume	Workers	2	3	VH	➤ Water jet to clean the surface of the pipeline to remove sludge or other deposits. ➤ Use appropriate gas detectors (i.e. explosion-proof design) to monitor the air in the confined space continuously until everyone has left the confined space ➤ Use blowers to direct fresh air through the air duct and remove air impurities ➤ Full body protective equipment (eg. face mask with cartridge filter, eye protector,) should be worn to all relevant workers to avoid inhale or contact with hazardous gas.	1	2	VL	E / F SS	CM / SA / GF / SO	1 / 5 / 8 / 11 / 12
		In-rush into the confined space of free flowing solid or liquid	Workers	2	3	VH	➤ Standby person should monitor the confined space continuously. If there is in-rush into the confined space of free flowing solid or liquid, notice Person-in-charge immediately for blockage. ➤ Prepare sufficient water pump to pump out water or flowing liquid. ➤ Under adverse weather (e.g. Heavy Rainstorm), evaluate whether it is possible to work in a confined space based on the on-site conditions at the time ➤ If the in-rush is not controllable, engage emergency procedure and evacuate the confined space. ➤ Suspend the current discharging system ➤ The switch of the discharging system should be tagged and locked. The key should be kept by RSS and contractor representative. They should release the lock until they ensure all workforce was left from the confined space.	1	2	VL	E / F SS	CM / SA / GF / SO	1 / 5 / 8 / 11 / 12

Risk Assessment - Method Statement for Concrete Remedial Works Inside Cavern

No	Task	Hazards	Related Parties	Likelihood	Severity	Risk Rating	Control Measures	Likelihood Residue	Severity Residue	Risk Residue	Action By	Monitor & Review By	PPE
		Fire / Explosion	Workers	2	3	VH	➤ Provided Fire Extinguisher ➤ All un-necessary flammable materials should be removed away from the works location. ➤ No smoking inside confined space. ➤ All electrical devices must be earthed. ➤ Safe working procedure should be implemented. ➤ Safety training and training about the emergency procedures for fire should be provided to relevant workers. ➤ Strictly comply with and implement the Hot-Work-Permit system.	1	2	VL	E / F SS	CM / SA / GF / SO	1 / 5 / 8 / 11 / 12
		Tripping	Workers	2	2	M	➤ Provide proper access ➤ Provide proper lighting ➤ Proper materials stacking should be enforced. ➤ Demarcated materials storing area with warning sign displayed. ➤ All un-necessary materials should be removed from the working area. ➤ Communicate with relevant parties about the material stacking position.	1	2	VL	E / F SS	CM / SA / GF / SO	1 / 5 / 8
10.	General	Heat stroke & Heat Exhaustion	Workers	2	3	VH	➤ Allow workers to take regular breaks or rotate to other locations within the shift to reduce exposure to the hot environment ➤ Provide drinkable water for worker ➤ Request frontline supervisor to pay attention to any report of workers suffering from symptoms of heat stroke and inform their senior whenever symptoms observed	1	2	VL	E / F SS	CM / SA / GF / SO	1 / 5 / 8
		Slip, Trip and Fall	Worker	2	2	H	➤ Housekeeping should be maintained ➤ Provide proper access ➤ Sufficient illumination should be provided for the confined space	1	2	VL	E / F SS	CM / SA / GF / SO	1 / 5 / 8

Risk Assessment - Method Statement for Concrete Remedial Works Inside Cavern

No	Task	Hazards	Related Parties	Likelihood	Severity	Risk Rating	Control Measures	Likelihood Residue	Severity Residue	Risk Residue	Action By	Monitor & Review By	PPE
		Back Injury Caused by Material Handling	Workers	2	2	M	➤ Training for proper manual handling posture should be provided to all relevant workers. ➤ Manual handling assessment should be conducted to assess the hazard. ➤ Re-arrange the work to prevent a single worker to perform the task. ➤ Equipment such as pulley should be provided in order to reduce the transport distance. ➤ A clear & unobstructed access should be maintained.	1	1	VL	SE / E / GF / F / SO / SS	CM / GF / SM / SO	1/ 5/ 8
		Falling Object	All person on site	3	2	H	➤ Make sure that all area near the working position should be fenced off with suitable warning notice displayed. ➤ Secure / remove loose parts. ➤ Keep the workplaces clean & tidy. ➤ Safe working procedure should be implemented. ➤ Safety training should be provided to relevant workers. ➤ Communicate with relevant parties about the operation procedures.	2	1	VL	SE / E / GF / F / SO / SS	CM / GF / SM / SO	1/ 5/ 8

Risk Assessment - Method Statement for Concrete Remedial Works Inside Cavern

Legend:

CSAJV

Senior Construction Manager	-	SCM
Construction Manager	-	CM
Construction Team Leader	-	CTL
Site Agent	-	SA
Engineer	-	E
General Foreman	-	GF
Foreman	-	F
Safety Manager	-	SM
Safety Officer	-	SO

Assistant Safety Officer	-	ASO
Safety Supervisor	-	SS
Competent Person	-	CP
Subcontractor's Personnel	-	S
Environmental Officer	-	EO
Diver	-	D
Mechanic	-	M
Electrician	-	EL
Operator	-	OP

Specific Personal Protection Equipment:

Item	PPE	Standard	Item	PPE	Standard
1	Safety Boot	EN345	11	Gas Monitor	NIOSH
2	Respiratory Mask	EN149:2001, N95 (General)	12	Blower	-
3	Breathing Apparatus	NFPA Respiratory Mask (NOISH/MSHA)	13	Eye Protector	US Standard ANSI Z87.1 (Safety Goggles) EU Standard EN166 (BS 2092) (General); AUS Standard AS/NZS1337 (AS1337) (General);
4	Safety Line	JIS / KS; BS3367 (Similar standard)	14	Leather Gloves	BS0072 or similar standard
5	Safety Helmet with (Y-type Chin Strip)	AS/NZ 801, ANSI Z89, CSAZ94, EN397, JIST8131, GB2811	15	Fall Arrestor	EN352(Fall arrestor or similar standard) EN354, EN355, EN14397 (Shock Absorber or similar standard)
6	First Aid Box	-	16	Face Shield	-
7	Full-Body Safety Harness	BSEN361 (Full-Body), BSEN360, EN353	17	Apron	-
8	Reflective Vest	EN471	18	Overall	-
9	Ear Protector	E-A-R Model 1000 (Ear Muffor similar standard), 3M 1110	19	Welding Goggle	EN169; BS679; EN175; BS 1542; DIN 58210; BS1729 DIN 58211 (General); AS/NZS1338 (AS1338) (Welding)
10	PU Gloves	EN388, EN420			

Remark: Detail of Personal Protection Equipment Refer to Construction Health and Safety Plan/ BS 8800 Risk Assessment

Risk Assessment - Method Statement for Concrete Remedial Works Inside Cavern

Table 1 : A simple risk estimator

表一：簡易風險級別評估

Likelihood of harm (see Table 2) 發生傷害的可能性	Severity of harm 傷害的嚴重性		
	Slight harm (1) 輕微傷害	Moderate harm (2) 中度傷害	Extreme harm (3) 極端傷害
Very unlikely (1) 非常不可能	VERY LOW RISK (VL) 極低風險	VERY LOW RISK (VL) 極低風險	HIGH RISK (H) 高風險
Unlikely (2) 不可能	VERY LOW RISK (VL) 極低風險	MEDIUM RISK (M) 中風險	VERY HIGH RISK (VH) 極高風險
Likely (3) 可能	LOW RISK (L) 低風險	HIGH RISK (H) 高風險	VERY HIGH RISK (VH) 極高風險
Very likely (4) 非常可能	LOW RISK (L) 低風險	VERY HIGH RISK (VH) 極高風險	VERY HIGH RISK (VH) 極高風險

Note These categorizations and the resulting asymmetry of the matrix arise from the examples of harm and likelihood illustrated within this British Standard. Organizations should adjust the design and size of the matrix to suit their needs.

由上述的傷害及可能性而得出的分類及不對稱矩陣，並不在英國標準限制內，各機構可調整及設計有關矩陣的大小以符合本身需要

Table 2 : A simple risk categorization

表二：簡易風險級別評估

Category of risk 風險類別	Evaluation of tolerability 可容忍程度的評估
Very Low 極低	Acceptable 可接受
Low* 低	Risks that should be reduced so that they are tolerable or acceptable. 可容忍或可接受的風險仍需降低
Medium* 中	
High* 高	
Very High 極高	Unacceptable 不可接受

*In this example these three categories are used to allow for different types actions or different timescales for action that might need to be applied according to the level risk.

*此 3 項風險類別例子可容許用不同的行動或時間表來執行降低有關風險

Risk Assessment - Method Statement for Concrete Remedial Works Inside Cavern

1. Type of Training / Toolbox Talk to be provided:

Manual handling, Safety use of PPE, Adverse Weather Safety, Working near underground Utilities Safety

Housekeeping, Confined Space Safety Briefing, Emergency Procedure, Excavation, Safety

2. Safety PPE / Equipment Required:



Safety Helmet



Ear protector



Goggle



Welding Glasses



Breathing Apparatus



Reflex Clothing



Safety Shoes



Gloves



Dust Mask



Safety Belt / Harness



Resuscitator



Other (Please Specify)

3. Risk Assessment Team Member

Name	Designation	Signature	Date
Yu Cho Yiu, Curtis	Assistant Engineer		25 April 2026
Hung Ling Fung, Samuel	Site Agent		25 April 2026
Yeung Lung Fung, Kenneth	Construction Manager		25 April 2026
Tsang Chuen Ming, Michael	Safety Manager		25 April 2026

Risk Assessment No. RA009

Method Statement for Concrete Remedial Works Inside Cavern Rev.0

Assessed by :


Tsang Chuen Ming,
Michael

Safety Manager


Yeung Lung Fung,
Kenneth

Construction Manager

Date :

25 April 2026

25 April 2026

Prepared by :


Yu Cho Yiu, Curtis
Assistant Engineer

Approved by :


Hung Ling Fung, Samuel
Site Agent

Date :

25 April 2026

Date :

25 April 2026

Issue Date :

25 April 2026

Next review date :

24 October 2026